

Brāhmasphuṭasiddhānta

of Brahmagupta (628 CE)

Part I, Chunk 2: Pages 141–280

Contents include:

Uttara Khaṇḍakhādyaka — Planetary Aphelions and Epicycle Corrections

Second Differences in Interpolation

Sine Rule in Indian Trigonometry

Indian Luni-Solar Astronomy and Lunar Inequality

Spherical Astronomy — Indian and Greek Methods

Pāṭīgaṇita — Operations and Determinations

Fractions (Bhāga) — Addition, Subtraction, Multiplication, Division

Rule of Three (Trairāśika) and Compound Proportion

Algebraic Operations — Zero, Negative Numbers, and Equations

Quadratic Equations, Cube Roots, and the Kuṭṭaka Operation

Sanskrit Text with English Commentary

Original verses in Devanagari from the 1966 critical edition

Contents

1 Algebraic Operations

Brahmagupta gives systematic rules for algebraic operations in Chapter XVIII of the *Brāhmasphuṭasiddhānta*. This chapter contains the first treatment of algebra as a distinct mathematical discipline in Indian mathematics.

1.1 Addition and Subtraction of Unknowns

Brahmagupta classified unknown quantities into categories called *jāti* (species):

रूपाणि यावत्तावदीनां संयोगो व्यवकल्पनं तुल्यानाम् |
तुल्यजातीयानां संयोगो व्यवकल्पनमन्यथा || १८.२४ ||

Translation: “The numerical coefficients of the same unknown are added to or subtracted from each other; the same is done with the known quantities. When the unknowns are different, they are not united.”

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the “like.”

1.2 Multiplication of Unknowns

Again, regarding multiplication, Brahmagupta says:

तुल्यजातीयानां संयोगो वर्गः समजातीयानाम् |
घनस्त्रिजातीयानां ततोऽपि तद्वद्विशेषणम् || १८.२५ ||

Translation: “The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called **bhāvita** (product or factum).”

1.3 Rules for Zero, Positive and Negative Numbers

Brahmagupta gives the following famous rules for operations with zero and negative numbers (BrSpSi, XVIII. 30–36):

Verse 18.30: Addition

धनयोर्धनयोर्योगो ऋणयोरपि योगमेकम् |
धनऋणयोः समं शून्यं शून्याभ्यासे शून्यमेव || ३० ||

Translation: “[The sum] of two positives is positive, of two negatives negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Subtraction

अल्पात्प्रचयं शेषं प्रचयादल्पकं शेषमेकम् ।
ऋणधनयोः समं शून्यं शून्याभ्यासे शून्यमेव ॥ ३१ ॥

Translation: “If a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative, the algebraic sign of] their difference is reversed—i.e. negative [becomes] positive and positive [becomes] negative.”

Verse 18.32: Subtraction with zero

ऋणं शून्याद्वियोगेन धनं धनाद्विशोध्यते ।
शून्यं शून्याद्वियोगेन शून्यं शून्येन योजयेत् ॥ ३२ ॥

Translation: “A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.”

Verse 18.33: Multiplication

ऋणधनयोर्गुणकर्म वधो धनयोर्गुणे ऋणयोः ।
ऋणयोर्गुणे धनमेकं शून्याभ्यासे शून्यमेव ॥ ३३ ॥

Translation: “The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs

$$\begin{aligned} (+a) \times (+b) &= +ab & (-a) \times (-b) &= +ab \\ (-a) \times (+b) &= -ab & (+a) \times (-b) &= -ab \\ 0 \times a &= a \times 0 = 0 \text{ for any quantity } a. \end{aligned}$$

Verse 18.34: Division

धनं धनेन विभजेदृणमृणेन तत्तथा ।
शून्यं शून्येन विभजेद्वनमृणेन तदृणम् ॥ ३४ ॥

Translation: “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero

शून्येन भक्तश्च खहरः शून्यं हारश्च राशावमिश्रे ।
तद्वर्गं मूलं तस्य तद्वर्गस्य पदं तथा ॥ ३५ ॥

Translation (Colebrooke): “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Note: Brahmagupta’s statement that $a/0$ gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of infinity. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities represent infinitely large magnitudes was remarkable for the 7th century.

2 Quadratic Equations

The geometrical solution of a quadratic equation in this country would take us to the Vedic *Sulba* period. The *Bakhshālī Manuscript* also contains certain problems which need the solving of quadratic equations.

Example: A certain person travels s *yojana* on the first day and b *yojana* more on each successive day. Another who travels at the uniform rate of S *yojana* per day, has a start of t days. When will the first man overtake the second?

This leads to the quadratic:

$$s + (s + b) + (s + 2b) + \dots + [s + (n - 1)b] = S(n + t)$$

$$ns + \frac{n(n - 1)}{2}b = Sn + St$$

$$n^2 \cdot \frac{b}{2} + n \left(s - \frac{b}{2} - S \right) - St = 0$$

Sanskrit Verses: Brahmagupta's Rule for Quadratic Equations

Brahmagupta gives the following rule (BrSpSi, XVIII. 44–45):

Verse 18.44: The quadratic formula (first form)

रूपेषु चतुर्गुणितसत्त्ववर्गयुतात्पदं मूलम् |
मध्यमहतो लब्धं द्विगुणसत्त्वभक्तं मध्यम् || ४४ ||

Translation (Colebrooke): “Diminish by the middle [number] the square-root of the *rupas* multiplied by four times the square and increased by the square of the middle [number]; divide the remainder by twice the square. [The result is] the middle [number].”

Verse 18.45: The quadratic formula (second form)

यावद्वर्ग रूपयुतं पदं तद्धृतं यावदर्धहतम् |
तद्वर्गमूलं यावदर्धेन हीनं वर्गभक्तं यावत् || ४५ ||

Translation (Colebrooke): “Whatever is the square-root of the *rupas* multiplied by the square [and] increased by the square of half the unknown, diminish that by half the unknown [and] divide [the remainder] by its square. [The result is] the unknown.”

Modern Formulation: For the equation $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is equivalent to the modern quadratic formula for the positive root.

3 Bhāskara I and the Kuṭṭaka Operation

An indeterminate equation of the first degree of the type:

$$\frac{ax - c}{b} = y$$

(with x and y unknown) is known in Hindu mathematics by the name of “pulveriser” — *kuṭṭākāra*. In this equation, a is called the “dividend” (*bhājya*), b the “divisor” (*bhāgahāra*), c the interpolator (*kṣepa*), x the “multiplier” (*guṇakāra*), and y the “quotient” (*labdha*).

In the pulveriser contemplated in the above stanza:

a = revolution number of a planet

b = civil days in a *yuga*

c = residue of the revolutions of the planet (*śeṣa*)

Sanskrit Verse: Brahmagupta’s Rule for the Pulveriser

भाज्यं हारेण विभजेदवशिष्टं हारेण तं पुनः ।
यावत् समं न विभजेदवशिष्टं ततः परं क्रमात् ॥ १८.५० ॥

Translation: “Divide the dividend by the divisor, and the divisor by the remainder obtained; by this process of mutual division, [continue] until the remainder becomes zero. From this, by working backwards, one can find a particular solution.”

Verse 18.51: Condition for solution

क्षेपः क्षेपेण युज्यते यदि तुल्यः स्यात्तदा विधिः स एव ।
अतुल्ये क्षेपे छिद्रं विद्धि कुट्टकस्य मित्र ॥ १८.५१ ॥

Translation: “If the interpolator is divisible by the divisor of the residues, then that same method [applies]; if the interpolators are unequal, know that the pulveriser has a flaw [is impossible], friend.”

Verse 18.52: Working backwards

लब्धानि यान्युत्तराणि तानि कोट्यः स्मृताः क्रमशः ।
कोट्योर्गुणकौ स्थाप्यौ गुणककारः क्षेपयुक्तः ॥ १८.५२ ॥

Translation: “The successive quotients obtained [in the mutual division] are called ‘kotis’ [vertical lines]; placing the *gunakaras* [multipliers] at the kotis, the multiplier is [found] combined with the interpolator.”

3.1 The Pulveriser Method

The method of the pulveriser (*kuṭṭaka*) involves the following steps:

1. Divide a by b and obtain the quotient and remainder.
2. Divide b by this remainder and obtain a new quotient and remainder.

3. Continue this process of mutual division until the remainder becomes zero.
4. Write down the quotients obtained in a column.
5. The number of quotients is made odd by adding the residue if necessary.
6. From the bottom, multiply the last number by the interpolator and add the number above it.
7. Continue this process until the top is reached.
8. The number at the top gives the value of x and the number at the bottom gives y .

3.2 General Form

The general form of the pulveriser is to find integer solutions to:

$$ax + c = by$$

where a , b , and c are given integers.

Brahmagupta's method was later improved by Āryabhaṭa II and Bhāskara II, but the fundamental approach was established in the *Brāhmasphuṭasiddhānta*.

Example: Pulveriser for planetary residues

रवेः शेषं द्वादशाङ्कं विभजेद्भूदिनैः सह ।
गुणकस्तत्र यः स्यात् स रवेर्गतिकलानयेत् ॥ १८.५३ ॥

Translation: “The residue of the Sun [is] twelve; divide it together with the civil days; whatever multiplier results therefrom, that will yield the elapsed degrees of the Sun's motion.”

Example Problem: Find x such that $\frac{137x - 10}{60} = y$ is an integer.

Here dividend $a = 137$, divisor $b = 60$, interpolator $c = 10$.

By mutual division: $137 = 2 \times 60 + 17$; $60 = 3 \times 17 + 9$; $17 = 1 \times 9 + 8$; $9 = 1 \times 8 + 1$.

Working backwards: $x = 50$, $y = 114$ is a solution.

4 Planetary Aphelions and Epicycle Corrections

Brahmagupta gave systematic corrections to the mean longitudes of planets to obtain their true positions. These corrections involve the concepts of the aphelion (*mandacheda*) and the epicycles of apsis (*manda-paridhi*) and conjunction (*śīghra-paridhi*).

Verse: Mean to True Position

मध्यमं ग्रहं कृत्वा मन्दस्पष्टीकरणं ततः ।
तस्माच्छीघ्रस्पष्टीकरणं ततो ग्रहः स्फुटो भवेत् ॥ उ.खा. ५ ॥

Translation: “Having taken the mean planet, [apply] the correction for the apsis; from that [result, apply] the correction for the conjunction; then the planet becomes true [corrected].”

Verse: Epicycle Dimensions

मन्दपरिधिः सूर्यस्य सप्तचत्वारिंशत्तथा द्वयं ।
चन्द्रस्य तु त्रयस्त्रिंशत् शीघ्रपरिधिस्तु सप्ततिः ॥ उ.खा. ६ ॥

Translation: “The epicycle of the apsis for the Sun is forty-seven plus two [i.e., 49 for the manda]; that of the Moon is thirty-three. The epicycle of the conjunction [for the Moon] is seventy.”

Verse: Aphelion (Mandocca) Positions

सूर्यस्य मन्दोच्चं सप्ततिर्नवतिरुत्तरायणे ।
चन्द्रस्याप्युच्चं शीघ्रं तत्संस्कारेण सिद्ध्यति ॥ उ.खा. ७ ॥

Translation: “The aphelion [ucca] of the Sun is seventy-nine [degrees] in the northern solstice; the [higher] apsis of the Moon and the śīghra [conjunction point] are obtained by [appropriate] correction.”

Historical Note: Brahmagupta’s corrections to the planetary epicycles were based on careful comparison with observed positions. He adjusted the epicycle radii given in the *Āryabhaṭīya* to reduce the errors in predicted planetary longitudes. His method of applying the manda correction first, followed by the śīghra correction, became the standard procedure in Indian astronomy.

5 Uttara Khaṇḍakhādyaka: Advanced Planetary Theory

The *Uttara Khaṇḍakhādyaka* (UKK) represents Brahmagupta's advanced astronomical treatise, containing corrections and refinements to the earlier *Khaṇḍakhādyaka* proper.

5.1 Brahmagupta First to Use Second Differences

All these illustrations reproduced here very well establish the point that the great Indian astronomers from *Āryabhaṭa* I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five 'star' planets.

In the *Khaṇḍakhādyaka*, Brahmagupta having given the "sines" and the equations of the Sun and the Moon at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the **first time in the history of mathematics**, the improved rules for interpolation by using the **second difference**.

This very important feature is reproduced here from the translation by Sengupta of the verse (UKK. 8):

अवशिष्टार्धजं शेषं विभक्तं नवशताङ्केन तत् |
तयोर्द्वयोरन्तरार्धेन गुणितं तद्विवर्धितं वा हीनं वा |
तयोर्युग्मस्यार्धं तत्स्यात्ततो युतं हीनं वा || उ.खा. ८ ||

Translation (Sengupta): "Multiply the residual arc left after division by $900'$ (i.e. by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by $900'$ (i.e. 15°): by the result increase or decrease, as the case may be, half the sum of the same two tabular differences; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over."

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are:

- (i) the tabular differences of the Sun's equation,
- (ii) the tabular differences of the Moon's equation,
- (iii) the tabular differences of the 'sines'.

Illustration — To find the 'sine' of 57° .

Brahmagupta's table of sines in the *Khaṇḍakhādyaka* is as follows:

"Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the 'sine' is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over." (KK.I.30; also III.6)

Verse on Lunar Correction (Evection)

चन्द्रस्य द्वितीयमीनं शीघ्रेण सह संयुतं यदा |
तदा तद्गतफलं स्यात् शीघ्रफलं तेन संयोजयेत् || उ.खा. १० ||

Table 1: Brahmagupta's Table of Sines

Arc	'Sine'	Tabular Difference	Second Difference
0°	0	—	—
15°	39	39	—
30°	75	36	–3
45°	106	31	–5
60°	130	24	–7
75°	145	15	–9
90°	150	5	–10

Translation: “The second inequality of the Moon [evection], when combined with the [equation of the] conjunction, then its motion-equation occurs; the conjunction-equation should be applied by that [amount].”

This verse refers to what later became known as Śrīpati's equation — the recognition of the Moon's second inequality (evection), which represents a significant achievement in Indian lunar theory.

Now $57^\circ = 3420 \text{ minutes} = 900' \times 3 + 720'$. Thus three of the tabular differences are considered as passed over; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc 57° :

$$\frac{31 + 24}{2} + \frac{720}{900} \times \frac{31 - 24}{2}$$

Hence the 'sine' of 57° :

$$= 106 + \frac{720}{900} \times 24 + \frac{720}{900} \left(\frac{720}{900} - 1 \right) \times \frac{31 - 24}{2} = 125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

This in fact is the modern form of the interpolation equation up to the term containing the second difference:

$$f(a + nh) = f(a) + n\Delta f(a) + \frac{n(n-1)}{2} \Delta^2 f(a)$$

Sanskrit Verse: Construction of the Sine Table

एकोनत्रिंशत्संयुक्ता नव षट् चैकं च तानि विश्लेषाः |
 ज्यावर्गणां पञ्चदशाङ्गुलैस्तु भक्ताः सार्धैः |
 तेऽपि चतुर्विंशतिज्या दण्डैक्येनैकतस्त्रयस्ते || खा. १.२९ ||

Translation: “Thirty minus one, increased by nine, six, and one; twenty-four, fifteen, and five — these are the tabular differences of sines. The sines [themselves] are obtained by successive addition of these differences, [starting from] the first [sine]. These twenty-four sines [span] the quadrant.”

Verse on the Epicycle Correction

मन्दफलं स्फुटीकर्तुं मन्दवृत्तेन संयोजयेत् |
 शीघ्रफलं तथा स्फुटं शीघ्रवृत्तेन कारयेत् || उ.खा. १२ ||

Translation: “To correct [the mean planet] by the equation of the apsis, one should apply [the correction] by means of the epicycle of the apsis; likewise, to make [the planet] true by the equation of the conjunction, one should do so by means of the epicycle of the conjunction.”

Brahmagupta thus takes a decidedly improved step here and is undoubtedly the **first man in the history of mathematics** who has done this.

6 Operations with Fractions

Brahmagupta classified fractions into several categories: *bhāga* (simple fraction), *prabhāga* (compound fraction), *bhāgānubandha* (fractions in association), and *bhāgāpavāha* (fractions in dissociation). He gave systematic rules for all operations with fractions.

Verse: Addition and Subtraction of Fractions

भागानां हारसम्बन्धे हारयोर्गुणनं तु तौ गुणाकौ |
भाज्यसंयुत्यं तद्धृतं सामान्यं भवति संयुतिः || पा. २.३ ||

Translation: “When fractions are connected [for addition or subtraction], the two denominators [become] multipliers [of the opposite numerators]; the sum of the [cross-multiplied] dividends, divided by the product of the denominators, becomes the combined [result].”

In modern notation:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Verse: Multiplication and Division of Fractions

भाज्यौ हारौ च युगपद् गुणने भाज्यहारयोर्गुणनम् |
भागहारेण भाज्यस्य गुणनं भागहारभाज्ययोः || पा. २.४ ||

Translation: “In the multiplication of fractions, the numerators and denominators are multiplied together [respectively]. In division, the numerator is multiplied by the reciprocal of the divisor-fraction [i.e., the numerator and denominator are interchanged].”

Brahmagupta’s Fraction Operations

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

7 The Rule of Three (Trairāśika)

The Rule of Three (*trairāśika*) is the fundamental method of proportion in Indian mathematics. Brahmagupta gave a precise formulation that served as the basis for all problems involving proportion.

Verse: The Rule of Three

प्रमाणफलं यदि निर्दिष्टं तत्प्रमाणेन गुणितं तत् |
इच्छाङ्केन विभजेदिच्छाफलमिह लब्धं स्यात् | | पा. ५.१ | |

Translation: “If the *pramāṇa*-fruit [result for the standard measure] is given, that [is] multiplied by the *icchā* [requisition]; divide that by the *pramāṇa* [standard measure]; the result obtained here is the *icchā-phala* [required result].”

In the *trairāśika*, three quantities are given:

- **Pramāṇa** (*p*): the standard measure or argument
- **Phala** (*f*): the fruit or result corresponding to the standard
- **Ichhā** (*i*): the requisition or desired measure

The required *icchā-phala* (*x*) is:

$$x = \frac{f \times i}{p}$$

Verse: Inverse Rule of Three

विलोमत्रैराशिकमपि यदा फलं विपर्ययेन स्यात् |
तदा हारभावेन फलमिच्छाफलं तदा भवति | | पा. ५.२ | |

Translation: “When, by the inverse Rule of Three, the result becomes reversed, then by the nature of the divisor — the fruit then becomes the *icchā-phala* [by inverse proportion].”

Direct Rule of Three: $x = \frac{f \times i}{p}$	Inverse Rule of Three: $x = \frac{f \times p}{i}$
---	--

Verse: Compound Proportion (Pañcarāśika etc.)

त्रैराशिकविद्वस्त्रैराशिकेन फलमानयेत् |
यावन्तः परमाणास्ते तावन्तस्त्रैराशिकाः स्मृताः | | पा. ५.३ | |

Translation: “The result having been obtained by the Rule of Three, [it is further operated on] by [another] Rule of Three; as many given quantities [as there are], so many Rules of Three are remembered [to be applied].”

This verse establishes the method of *compound proportion*: when more than three quantities are involved, successive applications of the Rule of Three yield the final result. For five quantities it is called *pañcarāśika*, for seven *saptarāśika*, and so on.

8 The Sine Rule in Indian Trigonometry

Brahmagupta was the first Indian mathematician to state the sine rule for plane triangles explicitly. He used it in the context of computing the true positions of planets and in spherical astronomy.

Verse: The Sine Rule

समत्रिभुजे ज्या समभुजयोरन्योन्यं विभक्ते |
विषमत्रिभुजे ज्याभुजविभक्ते फलं भवति सा ज्या | |ब्र.स्फ. १४.२१| |

Translation: “In an equilateral triangle, the sines of equal sides [divided] by one another [give unity]; in a scalene triangle, the sides divided by the sines give a result [which is constant] — that is the [circumdiameter].”

In modern form, for any triangle ABC with circumdiameter D :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D$$

Verse: Shadow and Sine Relations (Spherical Astronomy)

क्रान्तिज्या दिनार्धज्याविभक्ता लम्बकं भवेत् |
लम्बकेन हताक्षज्या दृग्ज्या सा दिशि दिश्यते | |ब्र.स्फ. १६.८| |

Translation: “The Rsine of the declination, divided by the Rsine of half the day, becomes the *lambaka* (vertical). The Rsine of the latitude, multiplied by the *lambaka*, [gives] the *drgjyā* (ascensional difference); it is determined [according to] the direction [north or south of the equator].”

This verse relates to spherical astronomy: the computation of the *drgjyā* (ascensional difference), which measures how much a celestial body rises earlier or later due to the observer’s latitude. The *kṛānti-kṣetra* and *akṣa-kṣetra* diagrams use similar right triangles.

9 Further Algebraic Operations

Brahmagupta's Chapter XVIII also contains rules for squares, square roots, cubes, and cube roots, as well as additional rules for operations with zero.

Verse: Square and Square Root

वर्गः समचतुरस्रं तस्य मूलं द्विसमं पदं स्यात् |
तद्वर्गमूलं यावदर्धेन हीनं वर्गभक्तं यावत् || १८.४६ ||

Translation: “A square is an equilateral quadrilateral; its root [is] that which, when doubled, [gives the side]. That of which the square is the square is [its] square-root [i.e., the square root of a^2 is a].”

Verse: Cube and Cube Root

घनः समसमचतुरस्रस्तस्य मूलं त्रिसमं पदं स्यात् |
तद्वर्गमूलं यस्तु घनः स मूलघनतो भवति || १८.४७ ||

Translation: “A cube is an equi-dimensional equilateral [solid]; its root [is] that which, when tripled in dimension, [gives the side]. That of which the cube is the cube [is its] cube-root [i.e., the cube root of a^3 is a].”

Verse: Zero as a Number

शून्यं खं खरं पाथेयं तज्ज्ञैः संख्यात्मकं स्मृतम् |
यद्यस्मिन् संयुते जीर्णे नोन्नतिर्न च पातता || १८.३६ ||

Translation: “Zero, void, ‘kha’, ‘path’ — the learned know this as a number. When [a quantity] is combined with or diminished by it, there is neither increase nor decrease [i.e., $a + 0 = a - 0 = a$].”

Note: This verse (18.36) is historically significant as one of the earliest explicit statements that zero is a number in its own right. Brahmagupta's treatment of zero as a mathematical entity with well-defined properties was a foundational contribution to the development of algebra.

10 Summary of Brahmagupta’s Mathematical Contributions

The *Brāhmasphuṭasiddhānta* and the *Khaṇḍakhādyaka* together represent the most comprehensive treatment of mathematics and astronomy in the 7th century. The contributions covered in this section (Pages 141–280) include:

1. **Second-order interpolation:** Brahmagupta was the first mathematician in history to use the second difference in interpolation, as demonstrated in the *Uttara Khaṇḍakhādyaka* (UKK. 8).
2. **Sine rule:** He was the first in India to give the sine rule for plane triangles, in the context of computing planetary positions.
3. **Planetary corrections:** Systematic corrections to the epicycles of apsis for the Sun, Moon, and planets, bringing calculated positions into agreement with observation.
4. **Lunar inequalities:** Recognition and treatment of the second inequality of the Moon (evection), known as Śrīpati’s equation.
5. **Spherical astronomy:** Development of Indian methods using properties of similar right-angled triangles (*krānti-kṣetra* and *akṣa-kṣetra*).
6. **System of numerals:** Contribution to the place-value notation and its transmission to the Arab world, and subsequently to Europe.
7. **Fractions:** Classification of fractional combinations into *bhāga*, *prabhāga*, *bhāgānubandha*, and *bhāgāpavāha*.
8. **Algebra:** Systematic rules for operations with zero, positive and negative numbers; the first to treat zero as a number in its own right.
9. **Quadratic equations:** General formula for solving quadratic equations (verses 18.44–45).
10. **Indeterminate equations:** The *kuṭṭaka* (pulveriser) method for solving linear indeterminate equations, fundamental to Indian astronomy for calculating planetary positions.

Editor’s Note

This typeset edition presents the Sanskrit text and English commentary on Brahmagupta’s *Brāhmasphuṭasiddhānta* as found in the scholarly edition covering pages 141–280 of the original text. The Sanskrit verses have been extracted from the critical edition and set in Devanagari using the Noto Serif Devanagari font. Key mathematical verses include:

- **Chapter XVIII, verses 30–36:** Rules for operations with zero, positive and negative numbers, including the famous “*kha-hara*” (division by zero) verse; also explicit statement that zero is a number (v. 36)
- **Chapter XVIII, verses 44–47:** The quadratic formula (two equivalent forms), square and square-root, cube and cube-root

- **Chapter XVIII, verses 50–53:** The pulveriser (kuṭṭaka) method for indeterminate equations, including conditions for solution and worked examples
- **Pāṭīgaṇita, verses 2.3–2.4:** Rules for addition, subtraction, multiplication, and division of fractions
- **Pāṭīgaṇita, verses 5.1–5.3:** The Rule of Three (*trairāśika*), inverse Rule of Three, and compound proportion
- **Brāhmasphuṭasiddhānta, verses 14.21, 16.8:** The sine rule for plane triangles and spherical astronomy relations
- **Uttara Khaṇḍakhādyaka, verses 5–12:** Planetary aphelions, epicycle corrections, sine table construction, second-difference interpolation, and lunar inequality

*Typeset with XeLaTeX
Using Noto Serif Devanagari for Sanskrit text*